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The Impact of Egg Consumption on Visual Cognitive Performance in the Nutrition, Vision, and Cognition in Health: Egg (IONHealth-Egg) Study

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Objectives: The purpose of this study is to evaluate the impact of the nutrients in eggs, specifically in the yolk, on the visual cognitive performance (VCP) in generally healthy older individuals.

Methods: Ninety-nine generally healthy men (36) and women (63) aged 50 to 75 years were assigned randomly to one of five dietary treatment groups: four egg whites; two whole regular eggs; two whole omega-3-fortified eggs; four egg yolks; and a no-egg control. Subjects were asked to maintain their usual dietary intake including the treatment modifications daily for 30 d. VCP was measured using the NeuroTracker™ CORE (NT) 3-Dimensional (3-D) software program

(15 training sessions) during the final 2 wk of the study. Extensive secondary information including body composition, cardiovascular measures, sleep and exercise patterns, and general readiness to perform were also collected. Participants recorded food intake during the first 2 wk of treatment and each day they trained on the NT. Food logs were analyzed using Nutribase software. Statistical analyses were performed in SPSS using repeated measures ANOVA including significant covariates when appropriate.

Results: The dietary intervention was successful at creating distinct dietary intake differences for choline, lutein, omega-3 fatty acids and cholesterol ($P < 0.01$) between groups and improving VSP in all groups (+37%, $P < 0.01$). Additionally, men had significantly higher VSP (NTmean = 1.02 ± 0.33) than women (0.88 ± 0.27 , $P = 0.02$), however, no dietary intervention differences were significant ($P > 0.05$).

Conclusions: Dietary and cognitive interventions were successful at altering dietary intake and improving cognitive performance respectively but no egg consumption pattern was superior in improving cognitive response.

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